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Visually imperceptible adversarial patch attacks

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ABSTRACT

The vulnerability of deep neural networks (DNNs) to adversarial examples has attracted more attention. Many algorithms have been proposed to craft powerful adversarial examples. However, most of these algorithms modified the global or local region of pixels without taking network explanations into account. Hence, the perturbations are redundant, which are easily detected by human eyes. In this paper, we propose a novel method to generate local region perturbations. The main idea is to find a contributing feature region (CFR) of an image by simulating the human attention mechanism and then add perturbations to CFR. Furthermore, a soft mask matrix is designed on the basis of an activation map to finely represent the contributions of each pixel in CFR. With this soft mask, we develop a new loss function with inverse temperature to search for optimal perturbations in CFR. Due to the network explanations, the perturbations added to CFR are more effective than those added to other regions. Extensive experiments conducted on CIFAR-10 and ILSVRC2012 demonstrate the effectiveness of the proposed method, including attack success rate, imperceptibility, and transferability.

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1. Introduction

The development of deep learning technology has promoted the successful application of deep neural networks (DNNs) in various fields, such as image classification (Krizhevsky et al., 2017; Simonyan and Zisserman, 2014), computer vision (He et al., 2016; Taigman et al., 2014), natural language processing (Devlin et al., 2018; Goldberg, 2017). In particular, convolutional neural networks (CNNs), as one typical DNN, have shown perfect performance in image classification. However, lots of evidence shows that CNNs are extremely vulnerable to *adversarial examples* (Szegedy et al., 2013). An adversarial example is crafted from a clean example by adding well-designed perturbations that are almost imperceptible to human eyes but can easily fool a classifier.

Recently, a variety of methods were proposed to craft adversarial examples, such as L-BFGS (Szegedy et al., 2013), FGSM (Goodfellow et al., 2014), I-FGSM (Kurakin et al., 2016b), M-DI^2-FGSM (Xie et al., 2019), PGD (Madry et al., 2017), and C&W (Carlini and Wagner, 2017). These methods perturbed all the pixels in an image. At the same time, some methods only perturbed several pixels or a region of the image, such as JSMA (Papernot et al., 2016b), One-pixel (Su et al., 2019), Adversarial Patch (Brown et al., 2017) and LaVAN (Karmon et al., 2018). However, since these techniques directly perturbed all pixels without considering high-level semantic information, too many redundant perturbations are introduced in the irrelevant regions, e.g., background. On the contrary, we attempt to generate more hidden adversarial examples with imperceptible adversarial patch, as shown in Fig. 1. We select a feature region that make a substantial contribution for classification to add perturbations. It will be more hidden to fool the classifier with fewer perturbations than previous methods (Deng and Zeng, 2019). This special region is referred to as a *contributing feature region* (CFR) in this paper.

Although some local perturbations were proposed like JSMA (Papernot et al., 2016b) and One-pixel attacks (Su et al., 2019), they did not take account of the correlation between the neighboring pixels, *i.e.*, the perturbed pixels may not form a continuous region. Meanwhile, some local continuous region attacks were proposed, which are named as patch attacks (Brown et al., 2017; Karmon et al., 2018) or sticker attacks (Evtimov et al., 2017). Our method also belongs to patch attacks. However, our method is different

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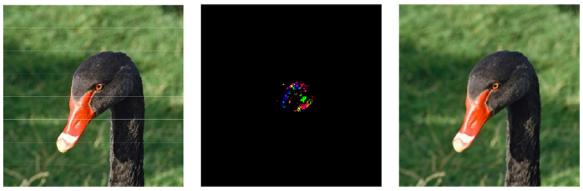


Fig. 1. Left: the original natural image predicted as black swan with 75% confidence. Middle: adversarial patch noise. Right: the adversarial image misclassified as crampfish with 70% confidence, in which we can not observe the adversarial patch.

from those existing patch attacks in three aspects. The first is that our CFRs shape is arbitrary, while the previous patches or stickers are generally regular. The second is that our CFR is located by the state-of-the-art explanation tools (Selvaraju et al., 2017; Zhou et al., 2016), while the location of patches or stickers is subjectively determined by the adversary. Finally, the perturbation magnitude of patches and stickers is not constrained, while our method limits the perturbations within a tiny bound to be imperceptible to human eyes. Recently, (Chen et al., 2021) proposed a novel DNN object contour attack named FineFool based on channel-spatial attention and pixel-spatial attention. Guo et al. (2021) proposed a local patch attack named ELLA that also uses model interpreters. Different from FineFool and ELLA, we use a soft mask and an optimization objective with an inverse temperature.

Furthermore, our work is different from image semantic segmentation (He et al., 2019) because our method pays more attention to the regions contributing to classification, while image segmentation is to find the object edge. In other words, we start with the network explanation while image segmentation focuses on the object itself. Recently, Xie et al. (2017) crafted adversarial examples to fool semantic segmentation and object detection, while Gu et al. (2020) leveraged the YOLO detector to locate regions sensitive to the added perturbations. Essentially, their methods did not fully utilize the network explanation. Besides, the size of our CFR is far smaller than the region obtained by object detectors. Wu et al. (2020) utilize the attention mechanism to investigate the transferability of adversarial examples across multi-models in black-box settings, while our goal is to improve the attack success rate with imperceptible patch perturbations in both white-box and black-box settings.

The main contributions can be summarized as follows:

- We propose a patch attack method based on arbitrary shape. The critical component in our method is a *soft mask* to represent an adversarial patch. The optimal patch position in the soft mask is located by the attention mechanism. The components of the soft mask are determined by the contribution of each pixel feature in the patch, which can make more refined optimization of adversarial perturbations than the traditional hard mask (0–1 matrix).
- Another critical improvement is that we propose a parameter named *inverse temperature* dividing loss function to make the model's output distribution of classes smoother, which can avoid the loss function approaching zero when the clean example is far away from the decision boundary. This technique guarantees the update of adversarial perturbations to maintain efficiency during the back-propagation.
- Extensive experiments are conducted on CIFAR-10 and ILSVRC2012, which demonstrate that CFR patch attack consistently outperforms state-of-the-art methods no matter in white-box or black-box settings. In brief, compared with the recently proposed patch attacks, the patch crafted by our method is imperceptible; among all the global attacks, the magnitude of our perturbations is the smallest.

The rest of the paper is organized as follows. The next section highlights the related works in various adversarial attacks, including global perturbations and local perturbations at pixel levels. Then we describe our method to craft perturbations on CFRs in Section 3, and the experimental results are shown in Section 4. Finally, we conclude the paper in Section 5.

2. Related work

In this section, we introduce some related work in adversarial attacks according to the perturbed regions. These attacks are divided into global adversarial attacks and local adversarial attacks. Our approach proposed in this paper belongs to the latter. However, different from previous local attacks at the pixel level, we add perturbations to some semantic regions through network explanations. The global adversarial attacks at the pixel level are introduced in Section 2.1, and the local adversarial attacks at the pixel level are presented in Section 2.2.

2.1. Global adversarial attacks at pixel levels

Szegedy et al. (2013) first found that DNNs are vulnerable to adversarial examples, and they proposed a box-constrained optimal perturbation method called L-BFGS. Since L-BFGS used an expensive linear search method to find the optimal perturbation, it was time-consuming and impractical. Goodfellow et al. (2014) proposed FGSM (Fast Gradient Sign Method) to generate adversarial examples. This method only performed a one-step gradient update along the direction of the sign of gradient at each pixel, so the computation cost was extremely lower. However, the generated adversarial example may not have the best performance since it is only a rough approximation. As shown in Kurakin et al. (2016b), a one-step attack like FGSM is easy to transfer across multi-models but also defensible. On the basis of FGSM, many other improved methods were proposed, such as I-FGSM (Kurakin et al., 2016a), in which iteration was applied to FGSM to generate adversarial examples. Tramèr et al. (2017) found that FGSM with adversarial training was more robust to white-box attacks than black-box attacks due to gradient masking. They proposed a RAND-FGSM, which added random noise when updating the adversarial examples to defeat adversarial training. Dong et al. (2018) proposed MI-FGSM, assuming that the gradients of each iteration were not only related to the current gradients but also related to the gradients of the previous iteration. More recently, M-DI^2-FGSM (Xie et al., 2019) based on MI-FGSM was proposed to improve transferability in black-box attacks.

Besides these FGSM series of attacks, a variety of other improved algorithms have been proposed. Moosavi-Dezfooli et al. (2016) proposed Deepfool that finds the closest distance from the original input to the decision boundary of adversarial examples. To overcome the non-linearity in the high-dimension space, they performed an iterative attack with a linear approximation. DeepFool provided less perturbation compared with FGSM. Carlini and Wagner (2017) proposed the C&W method to defeat defensive distillation. C&W considered three forms of perturbation constraints (ℓ_0, ℓ_1 and ℓ_∞ norm of the added perturbations) and adjusted the added perturbations with the optimization method. As far as we know, C&W is one of the most powerful attacks at the pixel level. All of these works achieve some significant progress, however, they do not fully take into account the semantics of images, which will bring more redundant perturbations to pixels.

2.2. Local adversarial attacks at pixel levels

Different from manipulating each image pixel for misclassification, several methods are proposed to perturb multiple pixels, which are called local adversarial attacks in this paper. A classic method is JSMA proposed by Papernot et al. (2016b) for targeted attacks. They perturbed a small number of pixels by a constant offset in each iteration step that maximizes the saliency map. However, JSMA has the disadvantage of over-modifying the value of the pixels, making the added perturbations easily perceived by the naked eye, and its adversarial strength is weak (Akhtar and Mian, 2018). Su et al. (2019) proposed the One-pixel attack method that successfully deceived the DNNs by modifying the value of a single pixel. Although this method is better for low-resolution images (such as CIFAR-10), the attack success rate for high-resolution images will be greatly reduced (such as ImageNet), and the cost is very large with the ℓ_1 distortion (Xu et al., 2018).

While the pixels perturbed by JSMA are usually nonadjacent, while in another method the pixels perturbed are in a continuous region. Evtimov et al. (2017) proposed a sticker attack that added noise patches as rectangular patterns on the top of trafic signs. Brown et al. (2017) presented a method called Adversarial Patch to create universal, robust, targeted adversarial image patches in the real world. Our method is different from them in three aspects. First, the size of the patch in Adversarial Patch is determined manually while in our method it is determined by CFR automatically. Second, the patch location of the Adversarial Patch can be random in the image, while the position in our method is located through Grad-CAM (Selvaraju et al., 2017) at the semantic level. Finally, the patches are recognizable in Adversarial Patch, while our patches are imperceptible. Karmon et al. (2018) suggested LaVAN that generates localized adversarial noises that cover only 2% of the pixels in the image, none of them cover the main object. Besides, it is transferable across images and locations, and can successfully fool a state-of-the-art Inception v3 model with very high success rates. Although these stickers and patch attacks have demonstrated powerful strength and can easily bypass existing defense approaches, there is no constraint on noise and they can be observed easily. Moreover, the location of stickers or patches is subjectively determined by the adversary. Gu et al. (2020) leveraged the YOLO detector to locate sensitive regions for perturbations. In contrast, the goal of our method is to locate a sensitive region through network explanations and add imperceptible perturbations.

3. Methodology

3.1. Preliminary

A CNN can be generally expressed as a mapping function $f(X,\theta):\mathbb{R}^m\to\mathbb{R}^c$, where $X\in\mathbb{R}^m$ is an input variable, θ denotes all the parameters and c is the number of classes. Typically, A CNN is comprised of convolutional layers with some method of periodic downsampling (either through pooling or stride convolutions). Let Z be the output vector of the penultimate layer, namely the Logits layer. This defines a mapping function: $X\mapsto Z$. The last layer of the CNN is a softmax layer. Then the softmax function can be expressed as $S(Z)_j=\exp(Z_j)/\sum_{i=1}^{c}\exp(Z_i)$, where Z_i is the $i-th$ element of Z, and $i\in[c]$, $[c]=\{1\dots c\}$ is a set of class labels. Thus the CNN can be expressed as $f(X)=S(W_sZ+b_s)$, where W_s and b_s are the weight matrix and bias vector of the softmax layer respectively. Given an input X with the ground-truth class label y, the predicted class label of X can be expressed as $\hat{y}=\arg\max_{i\in[c]}f(X)_i$. An adversarial example can be represented as $X'=X+\delta$, where X is a clean nature image, and δ is the perturbation. To obtain imperceptible perturbations, δ is always constrained by a p-norm, $\|\delta\|_p\le\epsilon$, where $p=0$, 2 or ∞ and ϵ is the perturbation bound.

3.2. Threat model

In general, the method used to generate adversarial examples needs some proper assumptions. These assumptions consist of a so-called threat model. Yuan et al. (2019) presented a deep learning threat model in two dimensions. The first dimension is the adversarial goal including *targeted attacks* and *untargeted attacks* according to the adversarial specificity. The second dimension is the attacker's ability defined by the amount of information that the attacker can obtain from target CNNs, which are divided into two categories, *i.e., white-box* attacks and *black-box* attacks.

Adversarial Goal: For untargeted attacks, the adversarial example X' satisfies $y'\ne y$, *i.e.*, y' can be any class label except y (the ground-truth label of X), where $y'=\arg\max_{i\in[c]}f(X')_i$. For targeted attacks, we specify a target class label y^*, and the adversarial example X' must satisfy $y^*=\arg\max_{i\in[c]}f(X')_i$ and $y^*\ne y$. In this paper, we mainly focus on untargeted attacks that are suitable for further adversarial training as a countermeasure.

Adversarial Capabilities are defined by the amount of information that the adversary has about the target classifier. The so-called white-box attack means that the adversary has almost all the information about the target CNN, including training data, activation functions, network topologies, and so on. The black-box attack, however, assumes that the attacker has no way to access the internal information of the pre-trained CNN, except the output of the model including the label and confidence. In this paper, we assume white-box settings as the same as PGD, C&W, etc. Nevertheless, all white-box attacks can be employed to lunch black-box attacks through a substitute model as described in Papernot et al. (2017).

3.3. Problem formulation

In this paper, we attempt to add perturbations to a local region instead of the whole image. To formalize the problem of this patch attack, let X_r denote the region to which the perturbation δ_r is added. Note that X_r is generally not a regular region. For the convenience of calculation, previous work generally introduces a 0–1 matrix M to represent the shape of X_r. Inspired by the views of Ilyas et al. (2019), we think the contribution of each feature (pixel) in X_r is also different. However, a traditional binary hard-mask cannot reflect the contribution of each feature, which only represents the shape of the region of interest. Therefore, we define this mask matrix in a soft way:

$$\tilde{M}(i,j)=\begin{cases}0<w\le 1 & X(i,j)\in X_r,\\ 0 & \text{otherwise}\ ,\end{cases}\tag{1}$$

where $X(i,j)$ is a pixel at the cell (i,j) of X and w is a weight measuring the contribution of $X(i,j)$. Thus we can transfer X_r to a matrix by $X\odot\tilde{M}$, where $\odot$ is the Hadamard product. Accordingly, δ_r can be represented by $\delta\odot\tilde{M}$, where δ is a global perturbation. Thus, obtaining an optimal δ_r can be modeled as the following constrained optimization problem:

$$\begin{aligned}\min\quad & \|\delta\odot\tilde{M}\|_p,\\ \text{s.t.}\quad & f(X+\delta\odot\tilde{M})\ne y,\\ & X+\delta\odot\tilde{M}\in[0,1]^m\,.\end{aligned}\tag{2}$$

However, solving the problem (2) is non-trivial. Instead, we obtain perturbations by maximizing the loss function as done in most previous work:

$$\begin{aligned}\max\quad & J\big(X+\delta\odot\tilde{M},y\big),\\ \text{s.t.}\quad & X+\delta\odot\tilde{M}\in[0,1]^m\,,\end{aligned}\tag{3}$$

where J is a loss function. Although Problem (3) is not fully equivalent to (2) and thus may not guarantee all obtained perturbations to flip the class label, its advantage is to fast find a possible perturbation within the constrained range like FGSM. We think that solving Problem (3) depends on two aspects. The one is to locate X_r, *i.e.*, identifying the soft-mask matrix $\tilde{M}$, as described in Section 3.4; the other is to find a proper loss function to solve Problem (3), as demonstrated in Section 3.5.

3.4. Construct soft-mask for CFRs

We use the "attention mechanism" to identify the position and feature contribution of a CFR. Suppose the input image $X \in \mathbb{R}^m$ is forward propagated through the CNN, and the final convolutional layer outputs the high-level feature map A of the image, where $A^{(k)} \in \mathbb{R}^{u \times v}$ represents the feature map of the $k-th$ convolutional kernel with the size of $u \times v$. Next, A passes through the fully connected layers and finally outputs a confidence vector Z. Let Z_y represent the logits of the $y-th$ class. A larger value of Z_y indicates X is predicted to the $y-th$ class with a greater probability. To this end, we compute the gradient of Z_y with respect to $A^{(k)}$, *i.e.*, $\partial Z_y / \partial A^{(k)}$ to measure the classification prediction importance of the $k-th$ convolutional kernel to the $y-th$ class. Furthermore, we adopt the global average pooling operation to calculate the weight $\lambda_y^{(k)}$ of the $k-th$ convolutional kernel:

$$\lambda_y^{(k)} = \frac{1}{u \times v} \sum_p \sum_q \frac{\partial Z_y}{\partial A_{pq}^{(k)}}, \tag{4}$$

where $A_{pq}^{(k)}$ is the activation at the cell (p, q) of the $k-th$ convolutional kernel. Thus, we obtain a feature activation map $\sum_k \lambda_y^{(k)} A^{(k)}$ for the $y-th$ class. Considering that only the positive elements in $\sum_k \lambda_y^{(k)} A^{(k)}$ have a positive effect on the classification, the result is further reactivated by ReLU to remove the influence of negative elements, and the final activation map of the $c-th$ class is obtained:

$$L_y = \mathrm{ReLU}\left(\sum_k \lambda_y^{(k)} A^{(k)}\right). \tag{5}$$

Suppose y is the ground-truth label of X, then for the example X, we can construct its soft-mask $\tilde{M}$ by L_y:

$$\tilde{M}(i, j) = \begin{cases} \frac{L_y(i,j)}{\sum_m \sum_n L_y(m,n)} & L_y(i, j) \geq \tau, L_y(m, n) \geq \tau, \\ 0 & \text{otherwise}, \end{cases} \tag{6}$$

where τ is a threshold. Since L_y is a heat map in essence, its normalized result is suitable as our soft-mask, which not only presents the shape of the CFR but also demonstrates the features' contribution.

3.5. Generate perturbations for CFRs

Previous work is merely to maximize the cross-entropy loss because they add perturbations to the whole image. In our work, we not only maximize the loss, but also minimize the perturbation. After locating CFR, we further generate the local perturbation δ_{CFR} on CFR. We design a new loss function to implement Problem (3), which consists of two parts: (1) a cross-entropy loss function J_{CE} for generating adversarial examples, and (2) an ℓ_2 regularization function to restrict the perturbation:

$$J = J_{CE} + \beta \frac{1}{\left\| \delta \odot \tilde{M} \right\|_2}, \tag{7}$$

where β is a hyper-parameter to control the degree of distortion (we set $\beta = 1$) and $\delta \odot \tilde{M}$ represents δ_{CFR}.

Remind the original cross-entropy loss function $J_{CE} = -\log S_y$, where $S_y = \exp(Z_y) / \sum_{i=1}^{c} \exp(Z_i)$. In adversarial settings, the adversary aims to maximize J_{CE} to obtain an adversarial example. However, when a clean example is far away from the decision boundary, its S_y tends to approach 1 and J_{CE} is close to 0 accordingly. In other words, this clean example has high confidence in its ground-truth class. Thus, the update of δ_{CFR} has minimal impact on J_{CE}, which is undesirable to us. To avoid this situation, we proposed a hyper-parameter T ($T > 0$) called *inverse temperature* inspired by the distillation idea Hinton et al. (2015). Then J_{CE} is modified as follows:

$$J'_{CE} = \frac{-\log(S_y)}{T}, \tag{8}$$

where $S_y \in (0, 1)$ and $\log(S_y) \in (-\infty, 0)$. If $0 < T < 1$, the lower bound of $\log(S_y)/T$ is magnified and $-\log(S_y)/T$ becomes larger, that is, J'_{CE} becomes larger. If $T > 1$, the lower bound of $\log(S_y)/T$ is reduced and $-\log(S_y)/T$ becomes smaller, that is, J'_{CE} gets smaller. Our goal is to maximize J'_{CE}, so we set $0 < T < 1$. Here, we notice T can smooth the confidence distribution of classes, which helps the update of δ_{CFR} to be maintained during the back-propagation. Thus we redefine Problem (3) as follows:

$$\begin{aligned} \max \quad & J'_{CE} + \beta \frac{1}{\left\| \delta \odot \tilde{M} \right\|_2}, \\ \text{s.t.} \quad & X + \delta \odot \tilde{M} \in [0, 1]^m. \end{aligned} \tag{9}$$

Finally, we use the hill climbing algorithm to solve problem (9) as shown in Algorithm 1.

Algorithm 1 Crafting Adversarial Patch Examples.

Input: A clean image (X, y), the iterations N, step size η, degree of distortion β, threshold τ, and inverse temperature T
Output: An adversarial example X'

initialize δ
// K is the number of feature maps in the last layer of convolution layers
$\lambda_y^{(k)} \leftarrow \frac{1}{u \times v} \sum_p \sum_q \partial Z_y / \partial A_{pq}^{(k)}$, $k = 1 \ldots K$
$L_y \leftarrow \mathrm{ReLU}\left(\sum_k \lambda_y^{(k)} A^{(k)}\right)$
// Get a CFR, $i = 0 \ldots u$, $j = 0 \ldots v$
if $L_y(i, j) \geq \tau$ **and** $L_y(m, n) \geq \tau$ **then**
    $\tilde{M}(i, j) \leftarrow L_y(i, j) / \sum_m \sum_n L_y(m, n)$
else
    $\tilde{M}(i, j) \leftarrow 0$
end if
for $t = 1 \ldots N$ **do**
    $J \leftarrow J_{CE}/T + \beta / \left\| \delta_t \odot \tilde{M} \right\|_2$
// Update δ
    $\delta_{t+1} \leftarrow \left(\delta_t + \nabla_{\delta_t} J \times \eta\right) \odot \tilde{M}$
    $X'_{t+1} \leftarrow \mathrm{Clip}(X + \delta_{t+1}, 0, 1)$
end for

4. Experiments

In this section, we first describe the datasets, models, and metrics used in our experiment. Then we show the impact of CFR on classification by simply setting CFR to 0. The results confirm that CFR plays a critical role in classification. For intuition, we visualize the adversarial examples and their perturbations of various methods. In white-box settings, we use ASR, SSIM, and ℓ_p to make a comparison between our method and the classic global attacks such as PGD and C&W, and local attacks such as JSMA, One-pixel.

In addition, two recent patch attacks Adversarial Patch and LaVAN are also compared with our method. Considering the above experiments are conducted on non-protected models, we further investigate the effect of our method on protected models. Furthermore, we compare the transferability of our method with other attacks in black-box settings. Finally, several key hyper-parameters are discussed in this paper.

4.1. Experiment setup

Datasets and Models. We validate our method on two benchmark datasets CIFAR-10 (Krizhevsky and Hinton, 2009) and ILSVRC2012 (Russakovsky et al., 2015). CIFAR-10 consists of 60,000 images with the size of 32×32, including 10 categories and each with 6000 images, in which 50,000 images are used for training and 10,000 images for tests. ILSVRC2012 contains 1000 categories, in which 1200 thousand images are used for training, and 50,000 images for tests. All the images we use to generate adversarial examples are correctly classified by all models, which can guarantee all the misclassified examples are adversarial examples. Two popular CNNs VGG (Simonyan and Zisserman, 2014) and ResNet (He et al., 2016) are selected for our experiment. According to their number of layers, they are further divided into VGG-11, VGG-13, VGG-16, ResNet-18, ResNet-34, and ResNet-50.

Evaluation Metrics. We use (1) the attack success rate (ASR) to measure the power of the adversarial examples, (2) the ℓ_p norm to measure the perturbation amplitude, and (3) the structural similarity (SSIM) index as a measurement of image similarity because human visual perception is highly sensitive to the structural information of an image (Wang et al., 2004).

(1) **ASR:** Given n clean images correctly classified by a CNN, the corresponding adversarial examples are obtained by a special generating method. Suppose X_i represents the $i-th$ clean image, its ground-truth label is y_i, and X_i' is its corresponding adversarial image. Then ASR can be obtained by the following formula:

$$ASR = 100 \times \frac{1}{n}\sum_{i=1}^{n}\mathbb{I}\left[\arg\max_{j\in[c]} f\left(X_i'\right)_j \neq y_i\right], \tag{10}$$

where $\mathbb{I}[\cdot]$ is an indicator function and $[c] = \{1 \ldots c\}$ is a set of class labels.

(2) **SSIM:** Given a clean image X and its corresponding adversarial image X', $\text{SSIM}\left(X, X'\right)$ measures the similarity between X and X'. A larger $\text{SSIM}\left(X, X'\right)$ indicates a higher similarity between the two images.

$$\text{SSIM}\left(X, X'\right) = \left[l\left(X, X'\right)\right]^{\alpha}\left[c\left(X, X'\right)\right]^{\beta}\left[s\left(X, X'\right)\right]^{\gamma}, \tag{11}$$

where α, β, $\gamma > 0$, $l\left(X, X'\right)$ is brightness comparison, $c\left(X, X'\right)$ is contrast comparison, and $s\left(X, X'\right)$ is structure comparison:

$$l\left(X, X'\right) = \frac{2\mu_X\mu_{X'} + c_1}{\mu_X^2 + \mu_{X'}^2 + c_1}, \tag{12}$$

$$c\left(X, X'\right) = \frac{2\sigma_{XX'} + c_2}{\sigma_X^2 + \sigma_{X'}^2 + c_2}, \tag{13}$$

$$s\left(X, X'\right) = \frac{\sigma_{XX'} + c_3}{\sigma_X\sigma_{X'} + c_3}, \tag{14}$$

where μ_X and $\mu_{X'}$ represent the average of X and X' respectively, σ_X and $\sigma_{X'}$ represent the standard deviation of X and X' respectively, $\sigma_{XX'}$ represents the covariance of X and X', and c_1, c_2, and c_3 are constants.

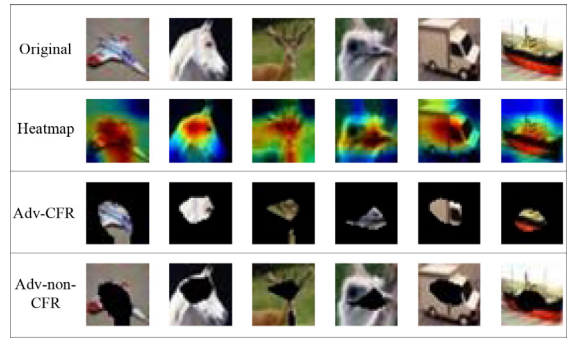


Fig. 2. Two groups of special adversarial images: Adv-CFR and Adv-non-CFR.

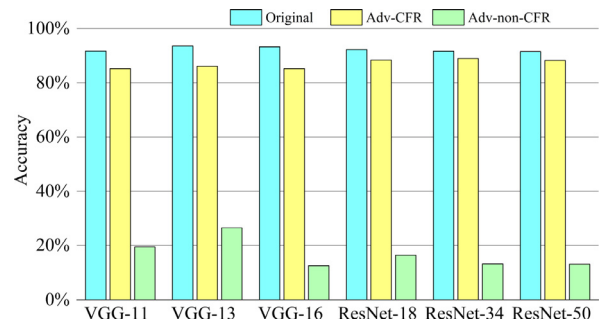


Fig. 3. Accuracy of different models with clean images, Adv-CFR, and Adv-non-CFR examples.

4.2. Impact of CFR on classification

We first evaluate the impact of CFR on the classifier through two groups of special adversarial images. The images in one group keep the pixels in CFR unchanged while the rest of the pixels are set to 0, which is denoted as Adv-CFR. On the contrary, the images in the other group keep the pixels unchanged other than CFR set to 0, which is denoted as Adv-non-CFR. These special adversarial images are crafted from 10,000 clean images on CIFAR-10 and the threshold $\tau = 0.2$ is adopted. Fig. 2 shows the examples of Adv-CFR and Adv-non-CFR. These examples of Adv-CFR and Adv-non-CFR are tested on VGG and ResNet, and the results are shown in Fig. 3. Compared to the original clean images, the accuracy of Adv-CFR decreases by no more than 3%, however, the accuracy of Adv-non-CFR decreases by at least 60%. The result shows that though the size of CFR is smaller than non-CFR, it yet plays a key role in classification. Therefore, the adversary modifying CFR is more effective than modifying other regions to fool a classifier.

4.3. Visualizing adversarial attacks

To intuitively illustrate the adversarial examples and their perturbations, we present them in Fig. 4, 5, and 6. We first compare our CFR patch with global pixel perturbations PGD and C&W on CIFAR-10 (see Fig. 4) and ILSVRC2012 (see Fig. 5). On CIFAR-10, the average SSIM of PGD is 0.94 and C&W is 0.96, while our method is 0.99. On ILSVRC2012, the average SSIM of PGD is 0.91 and C&W is 0.95, while our method is 0.99. Remind that a higher SSIM indicates a higher similarity between the images. It can be seen that the images with our patch perturbations are closer to the original clean images than those global perturbed images.

Two recent patch attack methods Adversarial Patch denoted as Adv.Patch (Brown et al., 2017) and LaVAN (Karmon et al., 2018) are compared with our method on ILSVRC2012. We do not conduct patch attacks on CIFAR-10 due to its low resolution and small size. For Adv.Patch, we use the same experimental setting as

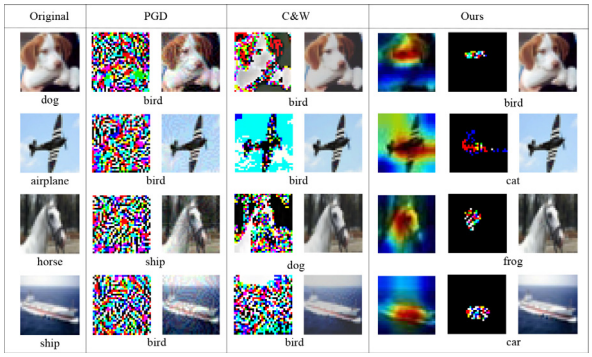


Fig. 4. Comparison of perturbations and adversarial examples generated by PGD, C&W, and our method on CIFAR-10 (all these three methods are constrained by $\ell_2 = 15$).

Brown et al. (2017) that the adversarial patch covers 10% of pixels. For LaVAN, which is an improved version of Adv.Patch, we follow the same implementation as Karmon et al. (2018) where the adversarial patch only covers 3% pixels. Fig. 6 presents the patches, adversarial examples, and SSIM of the above methods. Obviously, the patch crafted by Adv.Patch and LaVAN can be easily detected by humans. It can also be confirmed by SSIM that our method is ultimately close to 1, which is higher than Adv.Patch and LaVAN, and indicates the adversarial image crafted by our method is perfectly imperceptible to humans.

4.4. Comparison among adversarial attacks

To further evaluate the performance of our method, we report ASR, SSIM, and the ℓ_p distortion of different attack methods in Table 1. ResNet-18 and VGG-19 are leveraged to run on CIFAR-10 and ILSVRC2012 respectively. Six classic adversarial attack methods are compared with our method. For PGD, the perturbation bound $\epsilon = 16/255$, the step size $\alpha = 2/255$, and 20 iterations are adopted. For C&W, the constant $c = 1$, learning rate $lr = 0.01$ and 1000 iterations are adopted. Besides the global attacks PGD and C&W, two

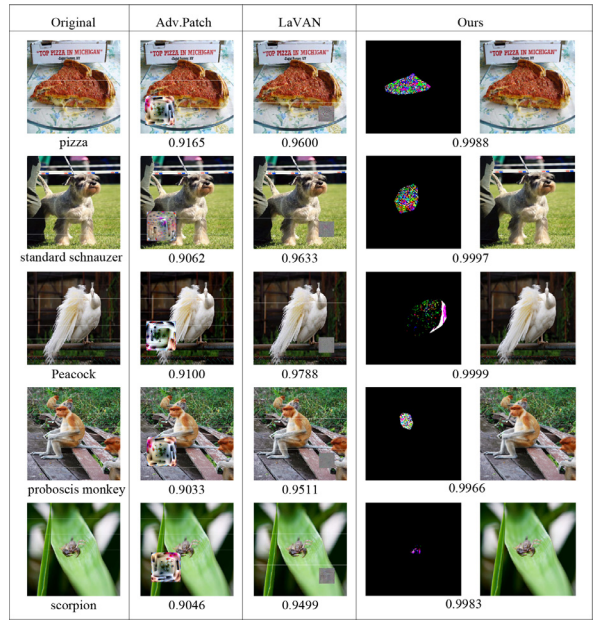


Fig. 6. SSIMs of three patch attacks on ILSVRC2012. The penultimate row visualizes the patch perturbation generated by our method.

local attacks JSMA and One-pixel attack are also considered. For JSMA, we set intensity variations $\theta = 0.3$. For the One-pixel attack, we adopt five pixel-modification.

It can be seen from Table 1 that our method outperforms other classic methods and recently proposed methods (ATA, FineFool, and ELAA). We achieve 99.89% ASR on CIFAR-10 and 99.80% ASR on ILSVRC2012, which exceeds the state-of-the-art C&W. We further analyze the distortion of adversarial examples with different methods. Among these methods, our method and One-pixel achieve the highest value of SSIM. It is no surprise that One-pixel performs well because its constraint ℓ_0 is only 15, which means only 5 pixels are modified. However, the number of modified pix-

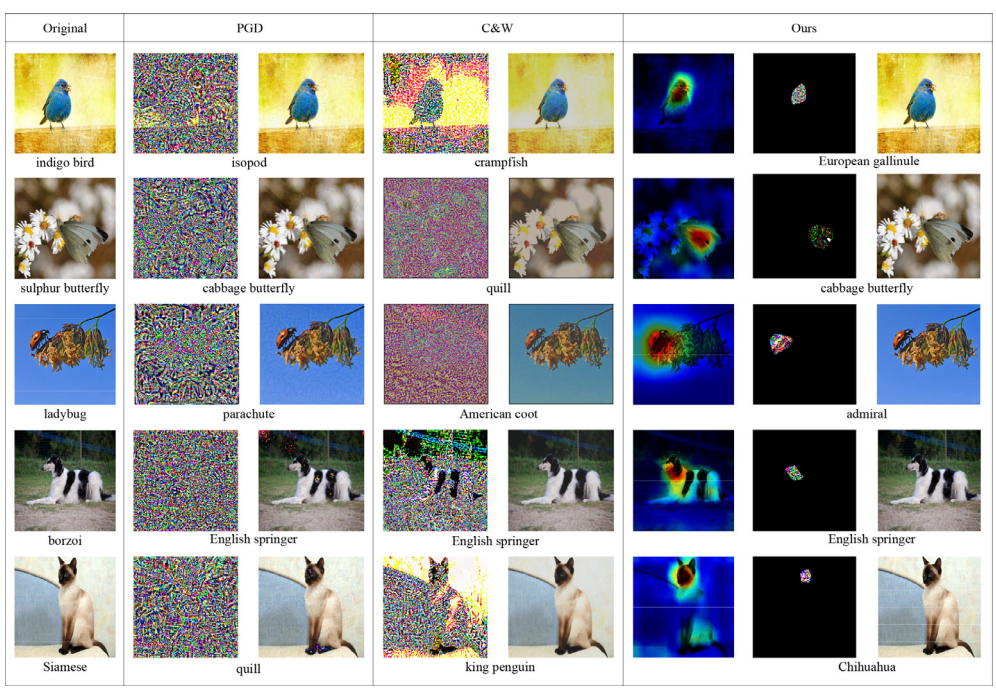


Fig. 5. Comparison of perturbations and adversarial examples generated by PGD, C&W, and our method on ILSVRC2012 (all these three methods are constrained by $\ell_2 = 15$).

Table 1
ASR, SSIM, and ℓ_p distortion of various attacks.

Dataset	Attack Methods	ASR	SSIM	ℓ_0	ℓ_1	ℓ_2	ℓ_∞
CIFAR-10	PGD	93.18%	0.94	3060	144.79	2.85	0.06
	C&W	97.44%	0.96	3072	18.11	0.49	0.05
	JSMA	90.33%	0.71	335	856.11	28.04	1.00
	One-pixel	80.77%	0.99	15	24.89	7.03	1.00
	ATA	99.78%	0.96	2760	128.97	2.61	0.08
	FineFool	98.51%	0.91	2992	144.28	2.83	0.08
	ELAA	98.79%	0.98	2073	38.93	0.74	0.09
	Ours	99.89%	0.99	2333	10.28	0.40	0.07
ILSVRC2012	PGD	97.70%	0.91	168,919	5,451.51	14.92	0.06
	C&W	99.33%	0.95	200,256	299	1.13	0.11
	JSMA	90.00%	0.94	447	75,375.33	194.35	1.00
	One-pixel	40.56%	0.99	15	29.43	8.69	1.00
	Adv.Patch	99.48%	0.90	14,700	8,223.06	70.57	1.00
	LaVAN	95.10%	0.96	7500	5,985.29	73.53	1.00
	ATA	99.35%	0.92	129,995	5375	16.18	0.06
	FineFool	97.99%	0.76	144,705	7,360.21	20.55	0.06
	ELAA	99.12%	0.99	34,469	1049.28	2.09	0.07
	Ours	99.80%	0.99	98,431	168	1.33	0.08

Table 2
Transferability (represented by the ASR on the target model) on CIFAR-10.

Model	Attack	VGG-11	VGG-13	VGG-16	ResNet-18	ResNet-34
VGG-11	PGD	93.51%	65.17%	63.46%	55.18%	58.15%
	M-DI^2-FGSM	99.10%	90.69%	**90.11%**	87.79%	85.09%
	C&W	93.94%	45.23%	46.46%	39.70%	49.25%
	ATA	99.73%	90.02%	90.08%	**88.23%**	85.11%
	FineFool	98.25%	89.99%	87.65%	85.72%	83.06%
	ELAA	99.43%	87.58%	86.23%	86.01%	81.79%
	Ours	**99.92%**	**91.00%**	86.97%	86.12%	**85.69%**
VGG-13	PGD	61.33%	95.81%	75.08%	61.36%	58.91%
	M-DI^2-FGSM	74.47%	97.70%	80.11%	79.78%	75.38%
	C&W	62.13%	95.72%	64.11%	61.95%	61.98%
	ATA	**81.06%**	99.03%	88.08%	80.09%	80.51%
	FineFool	78.21%	**99.11%**	90.00%	80.13%	79.07%
	ELAA	78.73%	97.43%	87.83%	80.09%	79.69%
	Ours	79.78%	98.80%	**90.39%**	**80.38%**	**80.68%**
VGG-16	PGD	56.57%	71.72%	90.43%	58.59%	56.78%
	M-DI^2-FGSM	66.77%	83.94%	90.79%	70.84%	69.38%
	C&W	51.52%	52.53%	96.97%	40.40%	48.48%
	ATA	81.26%	88.72%	95.91%	77.16%	75.49%
	FineFool	80.55%	86.39%	95.13%	72.96%	74.35%
	ELAA	80.01%	84.07%	93.99%	**78.32%**	74.84%
	Ours	**81.88%**	**88.29%**	**96.00%**	75.00%	**77.08%**
ResNet-18	PGD	60.70%	62.52%	60.92%	93.18%	64.22%
	M-DI^2-FGSM	72.83%	82.12%	83.42%	94.60%	84.35%
	C&W	61.28%	55.97%	54.88%	97.44%	56.24%
	ATA	85.00%	**85.01%**	83.37%	99.78%	86.72%
	FineFool	79.72%	78.75%	75.33%	98.51%	79.09%
	ELAA	83.48%	80.49%	77.92%	98.79%	88.91%
	Ours	**85.09%**	84.18%	**85.03%**	**99.89%**	**89.59%**
ResNet-34	PGD	60.28%	64.56%	60.92%	68.47%	92.72%
	M-DI^2-FGSM	78.08%	**87.78%**	88.89%	**91.39%**	98.60%
	C&W	57.75%	47.40%	46.00%	48.59%	90.34%
	ATA	84.66%	86.96%	88.08%	87.80%	96.75%
	FineFool	82.12%	87.43%	84.83%	81.27%	95.99%
	ELAA	81.93%	84.47%	87.57%	89.37%	97.07%
	Ours	**84.95%**	86.29%	**89.00%**	90.95%	**99.96%**

els of our method is far more than that of One-pixel and JSMA. It seems to contradict the value of SSIM achieved by our method. We will further explain the reason through ℓ_1, ℓ_2 and ℓ_∞ norms. As we know, ℓ_1 represents the sum of the absolute value of a perturbation on each pixel. In Table 1, ℓ_1 of ours is the smallest on CIFAR-10, and ranks only second to One-pixel on ILSVRC2012, which means our total perturbations are the smallest. Similarly, ℓ_2 and ℓ_∞ of ours are the smallest on CIFAR-10 and the second smallest on ILSVRC2012, where ℓ_2 measure the move distance of an adversarial example from its original example and ℓ_∞ means the largest change in the total pixels. As a result, these metrics finely explain the reason for the largest SSIM achieved by our method. In conclusion, our method is powerful (higher ASR), as well as imperceptible (higher SSIM).

4.5. Transferability

Transferability is an important property of adversarial examples, that is, the adversarial examples fooling one model can also fool other models (Papernot et al., 2016a). This property is used to investigate the black-box attack ability of adversarial examples. In this section, we compare the transferability of our CFR patch

Table 3
Transferability (ASR on the target model) on ILSVRC2012.

Model	Attack	VGG-16	VGG-19	ResNet-34	ResNet-50	ResNet-101
VGG-16	PGD	99.16%	84.74%	61.55%	60.92%	51.81%
	M-DI^2-FGSM	99.72%	85.90%	60.46%	62.11%	52.58%
	C&W	95.81%	80.44%	61.55%	61.45%	59.56%
	Adv.Patch	98.60%	35.78%	27.50%	24.40%	23.00%
	LaVAN	94.41%	55.48%	37.00%	27.70%	27.00%
	ATA	99.08%	85.71%	68.82%	65.63%	**64.91**%
	FineFool	99.02%	82.43%	63.18%	60.72%	58.23%
	ELAA	99.68%	85.82%	65.41%	**69.74**%	61.05%
	Ours	**99.92%**	**86.15%**	**67.12%**	69.47%	64.08%
VGG-19	PGD	86.73%	97.70%	61.14%	57.63%	54.52%
	M-DI^2-FGSM	88.41%	99.61%	56.11%	63.95%	54.13%
	C&W	77.37%	99.33%	60.46%	60.26%	58.40%
	Adv.Patch	69.27%	93.03%	27.40%	24.50%	23.00%
	LaVAN	73.46%	94.87%	35.00%	35.60%	30.00%
	ATA	89.72%	99.35%	70.74%	63.52%	63.73%
	FineFool	85.33%	97.99%	61.84%	60.27%	57.53%
	ELAA	**89.79%**	99.12%	63.73%	**68.31%**	61.41%
	Ours	83.90%	**99.80%**	**73.51%**	67.89%	**66.67%**
ResNet-34	PGD	75.70%	75.87%	99.18%	68.68%	58.27%
	M-DI^2-FGSM	70.39%	76.20%	**99.50%**	82.37%	75.84%
	C&W	74.16%	74.90%	92.53%	61.71%	58.40%
	Adv.Patch	19.47%	13.30%	98.31%	23.21%	37.04%
	LaVAN	24.44%	16.92%	96.88%	54.47%	46.90%
	ATA	77.41%	**77.43%**	99.12%	82.26%	81.37%
	FineFool	72.11%	72.89%	98.93%	78.48%	78.83%
	ELAA	**79.32%**	70.01%	99.25%	81.82%	81.73%
	Ours	75.92%	74.86%	**99.50%**	**82.84%**	**83.33%**
ResNet-50	PGD	**75.70%**	70.60%	62.36%	99.47%	60.72%
	M-DI^2-FGSM	68.44%	69.35%	**72.55%**	99.34%	84.37%
	C&W	74.58%	75.59%	61.82%	94.08%	57.75%
	Adv.Patch	19.41%	20.67%	40.00%	98.63%	44.06%
	LaVAN	24.58%	26.07%	46.74%	94.74%	51.03%
	ATA	75.12%	**76.33%**	70.01%	99.10%	85.41%
	FineFool	72.51%	68.72%	66.93%	98.65%	80.86%
	ELAA	75.24%	70.83%	65.72%	99.33%	80.94%
	Ours	70.68%	69.64%	71.77%	**99.74%**	**85.56%**
ResNet-101	PGD	74.30%	68.65%	60.19%	68.03%	99.48%
	M-DI^2-FGSM	61.87%	57.42%	69.84%	**85.39**%	99.61%
	C&W	**77.51**%	74.90%	62.91%	61.05%	94.19%
	Adv.Patch	20.26%	28.30%	44.05%	60.09%	98.13%
	LaVAN	26.20%	31.31%	52.31%	68.82%	94.19%
	ATA	77.11%	73.02%	79.24%	**86.73%**	99.72%
	FineFool	73.58%	68.41%	75.36%	78.75%	99.13%
	ELAA	76.19%	71.46%	76.61%	85.92%	99.73%
	Ours	76.87%	**76.06%**	**80.41%**	81.13%	**99.90%**

examples with the other classic adversarial examples in black-box settings. In these methods, PGD, M-DI^2-FGSM, and C&W are global adversarial attacks, while Adv.Patch, LaVAN, ATA, FineFool, and ELAA are local adversarial attacks. Note that the model for crafting adversarial examples is referred to as a *substitute model* (the first row of Table 2 and 3), while the model for testing adversarial examples is termed as a *target model* (the first column of Table 2 and 3) in this paper. When the substitute model is consistent with the target model, *i.e.*, the case in the diagonal of Table, it is equivalent to a white-box attack. In this section, we focus on transferability and merely consider the case that the target model is different from the substitute model.

On CIFAR-10, ASR on the target models is shown in Table 2, in which a higher value indicates higher transferability. For example, the adversarial examples generated by our method on the substitute model VGG-16, have the highest ASR on other target models, which indicates our method has higher transferability than other attack methods, including state-of-the-art M-DI^2-FGSM. Besides VGG-16, on the other substitute models, the adversarial examples generated by our method have the highest transferability in most cases. On ILSVRC2012, the similar conclusion is obtained (as shown in Table 3). In summary, our patch perturbation crafted with network explanations has a more powerful attack ability in black-box settings. We speculate the possible reason is that different classifiers share the similar CFR of an object, which is consistent with the result shown in Wu et al. (2020).

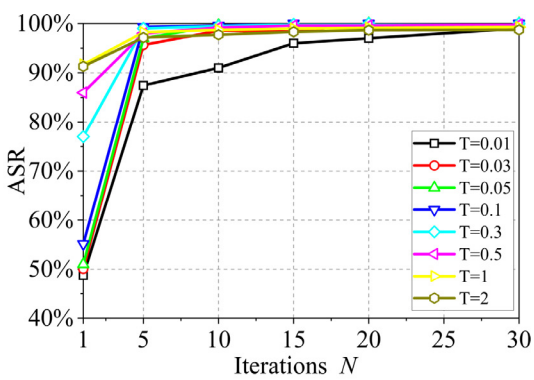
4.6. Adversarial attacks on protected models

The evaluation in Section 4.2–4.5 is conducted on the target model without protection. Now we further test the attack ability of our patch adversarial examples on protected models. We mainly focus on the model protected by adversarial training, because adversarial training is popularly considered as one of the most effective defenses (Shafahi et al., 2019). Here we use Fast adversarial training (Wong et al., 2020) and PGD adversarial training (Madry et al., 2017). For Fast adversarial training, we set the perturbation bound $\epsilon = 8/255$, step size $\alpha = 10/255$, and $epoch = 20$ on CIFAR-10. Meanwhile, we set the perturbation bound $\epsilon = 2/255$, step size $\alpha = 2.5/255$, and $epoch = 15$ on ILSVRC2012 respectively. For PGD adversarial training, we set 7 iteration steps, the step size $\alpha = 2/255$, total perturbation bound $\epsilon = 8/255$. Finally, we obtain four protected models ResNet-18-Fast, ResNet-18-PGD, VGG-16-Fast, and VGG-16-PGD as target models.

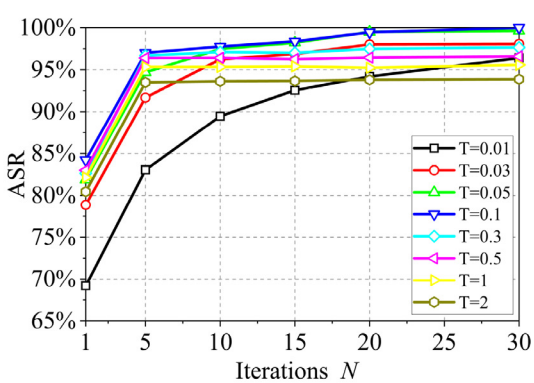
For CIFAR-10, we compare our method with PGD, C&W, ATA, FineFool, and ELAA. For ILSVRC2012, we add two patch attacks

Table 4
ASR of various attacks on protected models.

Dataset	Protected Models	Attack Methods	ASR
CIFAR-10	ResNet-18-Fast	PGD	63.66%
		C&W	70.79%
		ATA	79.15%
		FineFool	69.84%
		ELAA	75.22%
		Ours	79.00%
	ResNet-18-PGD	PGD	57.57%
		C&W	65.71%
		ATA	76.58%
		FineFool	65.69%
		ELAA	73.37%
		Ours	77.39%
ILSVRC2012	VGG-16-Fast	PGD	65.99%
		C&W	66.11%
		Adv.Patch	63.89%
		LaVAN	67.11%
		ATA	76.57%
		FineFool	67.96%
		ELAA	71.85%
		Ours	79.87%
	VGG-16-PGD	PGD	63.50%
		C&W	57.72%
		Adv.Patch	75.75%
		LaVAN	65.00%
		ATA	74.55%
		FineFool	70.52%
		ELAA	73.36%
		Ours	77.60%

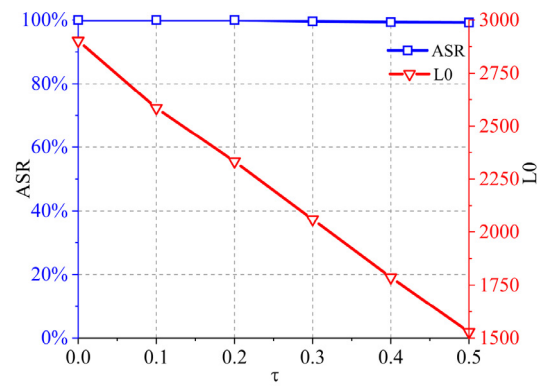


(a) ResNet-18 network on the CIFAR-10 ($\tau = 0.2$, $\eta = 10, \beta = 1$).

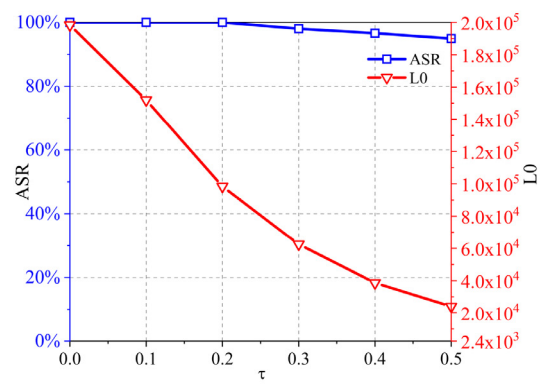


(b) VGG-16 network on the ILSVRC2012 ($\tau = 0.2$, $\eta = 20, \beta = 1$).

Fig. 7. The effect of iterations N and inverse temperature T on the ASR.



(a) ResNet-18 network on the CIFAR-10 ($N = 30$, $T = 0.1, \eta = 10, \beta = 1$).



(b) VGG-16 network on ILSVRC2012 ($N = 30$, $T = 0.1, \eta = 20, \beta = 1$).

Fig. 8. The influence of the threshold τ on the ASR and the ℓ_0 norm of perturbations.

Adv.Patch and LaVAN. Table 4 reports the results of the protected models under various attacks. We observe that adversarial training cannot achieve perfect performance against these attacks (ASR is higher than 50.00%). Among them, our CFR patch attack outperforms other attacks in ASR. For example, our method can achieve 79.87% ASR against VGG-16-Fast on ILSVRC2012.

4.7. Analysis of hyper-parameters

Iterations N and inverse temperature T are two dominant hyper-parameters in our algorithm, and here we investigate their effects on ASR. We observe that ASR tends to increase along with iterations N in Fig. 7. When N=30, ASR of our method can almost reach 100% on both datasets with a proper inverse temperature (e.g. T=0.1), which indicates that our objective function can find the global optimal adversarial examples with fine-tuned parameters. We further discuss the impact of inverse temperature T. As shown in Fig. 7, when $T > 1$ or T is too small, it will prevent our patch attack from achieving a higher ASR regardless of increasing iterations. Remind that the purpose of inverse temperature T is to prevent the loss J'_{CE} from decreasing to 0 as shown in Section 3.5. Nevertheless, when $T > 1$, J'_{CE} becomes smaller, which leads to a smaller ASR, e.g., when $T = 2$, it achieves the lowest ASR. Similarly, the smaller T makes J'_{CE} become so large that it deviates far from the original value and the optimal direction, which makes it hard to converge to the optimal solution. For instance, when $T < 0.1$, the ASR of patch attacks presents a downward trend. In summary, a moderate value of T is desirable, e.g., $T = 0.1$ for ILSVRC2012.

Threshold τ is also a dominant hyper-parameter to determines the shape of CFRs, that is, the size of the range of added perturbations. We use ℓ_0 norm to measure the number of perturbed pixels. Specifically, $\tau = 0$ means all pixels in the image are perturbed. As shown in Fig. 8, ASR and ℓ_0 are presented in the same figure with two independent vertical axes. We observe that increasing the threshold τ can decrease ℓ_0 norm, *i.e.*, the size of perturbed regions is reduced, however, it does not affect ASR of our patch attacks. The reason is that the most contributing pixels are maintained all the time despite the size of perturbed regions changed with τ. In other words, it reconfirms that the classification is mainly determined by the most contributing pixels.

5. Conclusions

Extensive experiments conducted on CIFAR-10 and ILSVRC2012 show that our patch attack outperforms existing global region attacks and local region attacks. Specifically, our patch attack has a higher attack success rate in both white-box and black-box settings. The main reason is that CFR plays a critical role in classification, and CFR of the same object is shared by multiple classification models. In addition, our crafted perturbations on CFR are imperceptible to human eyes. This imperceptible property is very attractive, which can be finely extended to other fields. For example, in the future, we will utilize this imperceptibility to explore new adversarial patch attack on medical image classification or segmentation, which requires very tiny perturbations to evade medical specialists. Meanwhile, we are going to develop new countermeasures against the CFR patches, such as smoothing CFR to filter adversarial perturbations.

Credit Author Statement

Yaguan Qian, Management and coordination responsibility for the research activity planning and execution. Development or design of methodology; creation of models. Acquisition of the financial support for the project leading to this publication. Preparation, creation, and presentation of the published work.

Jiamin Wang, Designing computer programs; implementation of the computer code and supporting algorithms, specifically performing the experiments.

Haijiang Wang, Conducting a research and investigationocess.

Zhaoquan Gu, Management and coordination responsibility for the research activity planning and execution.

Bin Wang, Preparation, creation, and presentation of the published work, specifically data presentation. Acquisition of the financial support for the project leading to this publication.

Shaoning Zeng, Provision of study materials, computing resources, or other analysis tools.

Wassim Swaileh, Preparation of the published work including pre- or post-publicationages.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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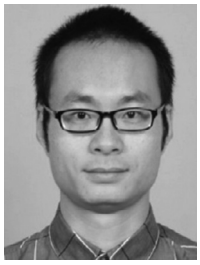

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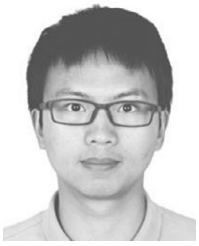

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